

MUNICIPAL COMPLAINT MANAGEMENT SYSTEM

Requirements Report

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1. Introduction

1.1 Project Title

Municipal Complaint Management System (MCMS)

1.2 Purpose

The Municipal Complaint Management System is a web-based application designed to provide an efficient and transparent platform for citizens to report problems related to municipal services and for municipal authorities to manage, assign, monitor, and resolve those complaints. The system supports complaint tracking, notifications, feedback, reporting, and selected AI-assisted features such as automatic complaint classification and priority prediction.

2. Problem Statement

Traditional municipal complaint systems often rely on physical complaint offices, telephone calls, emails, or disconnected online systems. These approaches can make it difficult for citizens to track complaints and for authorities to efficiently manage large numbers of complaints.

Lack of centralized complaint management.

Difficulty tracking complaint status.

Delayed response to complaints.

Complaints being assigned to the wrong department.

Lack of accountability for field workers.

Difficulty identifying overdue complaints.

Lack of evidence regarding complaint resolution.

Poor communication between citizens and municipal authorities.

Difficulty generating performance reports.

Duplicate complaints regarding the same civic issue.

Therefore, a centralized Municipal Complaint Management System is proposed to improve transparency, accountability, communication, and efficiency.

3. Data Collection

The system collects the information required to register, process, track, resolve, and evaluate municipal complaints. This section identifies the data sources, data items, collection methods, and purposes supported by the requirements of the proposed system.

3.1 Data Sources

Citizens provide registration/profile information, complaint details, locations, supporting evidence, and post-resolution feedback.

Municipal officers provide complaint verification/rejection decisions, department and worker assignments, priorities, deadlines, resolution verification, and escalation actions.

Field workers provide progress updates and resolution evidence for assigned complaints.

Administrators maintain users, departments, complaint categories, system settings, and related management information.

3.2 Data Collected

Data Category	Examples / Purpose
User Data	Name, email, phone number, address, password, and role for account and access management.
Complaint Data	Complaint ID, title, description, category, location, status, priority, deadline, and timestamps.
Evidence Data	Uploaded photographs/evidence, file information, uploader, and upload date.
Assignment Data	Complaint, department, field worker, priority, deadline, assignment date, and assigning officer.
Status History	Previous status, new status, person making the change, and change timestamp.
Notification Data	Notification ID, recipient/user, complaint reference, message, type, and creation time.
Feedback Data	Rating, comments, complaint reference, citizen, and creation date.
AI-Related Data	Complaint text and related complaint information used to generate advisory category, priority, or duplicate-detection suggestions.

3.3 Data Collection Methods

Web forms for registration, profile information, complaint submission, and feedback.

Location entry using a manual address, map-based location, or GPS-based location.

Image/evidence upload through supported file-upload functionality.

Status and progress updates entered by authorized municipal officers and field workers.

System-generated records such as complaint IDs, timestamps, status history, assignments, and notifications.

3.4 Data Use

To process and route complaints to appropriate municipal departments.

To assign complaints to field workers and monitor resolution.

To allow citizens to track complaint status and history.

To identify overdue complaints and support escalation.

To generate dashboards and reports.

To provide advisory AI classification, priority prediction, and duplicate complaint detection.

4. Requirement Elicitation Methods

Requirement elicitation is used to understand the needs, expectations, problems, and constraints of the people who will use or manage the Municipal Complaint Management System. The following methods are proposed for collecting and validating requirements.

Method	Stakeholders	Purpose
Questionnaire / Survey	Citizens, officers, field workers	Collect structured feedback from a larger number of participants about current problems, desired features, usability and priorities.
Interviews	Municipal officers, administrators, field workers	Obtain detailed information about existing workflows, responsibilities, verification, assignment, escalation and reporting.
Observation	Officers and field workers	Understand how complaints are currently received, verified, assigned, updated and closed in practice.

Method	Stakeholders	Purpose
Document Analysis	Municipal administration	Study existing complaint forms, registers, policies, categories, deadlines and reports to identify required data and rules.
Focus Group / Group Discussion	Mixed stakeholders	Discuss common issues and compare different stakeholder expectations.
Prototyping	Citizens and municipal staff	Show early screens/workflows and collect feedback before finalizing requirements.

4.1 Questionnaire / Survey

A questionnaire can be used to gather structured requirements from citizens and municipal staff. The questions should be short, clear, and focused on the complaint process.

No.	Sample Question
1	How do you currently report a municipal complaint?
2	What problems do you face while submitting a complaint?
3	Which complaint categories should be available in the system?
4	How important is the ability to track complaint status online?
5	Should citizens be able to upload photographs as evidence?
6	How should citizens receive complaint updates and notifications?
7	What information should officers see when managing a complaint?
8	How important is automatic routing to the correct department?
9	Should the system identify potentially duplicate complaints?
10	How useful would AI-based category and priority suggestions be?
11	What difficulties do field workers face while updating complaint progress?
12	What reports or dashboards should municipal management receive?

4.2 Requirement Validation

Collected requirements should be reviewed with stakeholders to remove ambiguity, identify conflicting expectations, prioritize important features, and confirm that the final requirements are feasible and testable.

5. Objectives

Provide citizens with an easy platform for registering municipal complaints.

Provide each complaint with a unique complaint ID.

Allow citizens to track complaint status.

Categorize complaints according to municipal services.

Route complaints to appropriate municipal departments.

Allow municipal officers to verify and manage complaints.

Allow officers to assign complaints to field workers.

Allow field workers to update complaint progress.

Allow workers to upload proof of completed work.

Allow officers to verify complaint resolution.

Notify citizens about important status changes.

Collect citizen feedback after resolution.

Identify overdue complaints and support escalation.

Provide dashboards and reports for municipal authorities.

Use AI to assist with complaint classification and priority prediction.

Improve transparency and accountability in municipal service delivery.

6. Scope of the System

6.1 Citizen Scope

Register an account

Login/logout

Manage profile

Submit complaints

Select complaint categories

Provide descriptions and location

Upload photographs/evidence

Track complaint status

View complaint history

Receive notifications

Request reopening of unresolved issues

Provide ratings and feedback

6.2 Municipal Officer Scope

View, search and filter complaints

Verify complaints

Reject invalid complaints

Assign departments and field workers

Set or modify priority

Set resolution deadlines

Monitor progress

Verify resolution evidence

Reopen complaints

Escalate overdue complaints

Generate reports

6.3 Field Worker Scope

Login

View assigned complaints

- View complaint details and location
- Update work progress
- Upload before/after images
- Mark assigned complaints completed
- View pending and completed assignments

6.4 Administrator Scope

- Manage users
- Manage officers
- Manage field workers
- Manage departments
- Manage complaint categories
- Manage system settings
- Monitor system activity
- Generate overall reports

7. Stakeholders

Stakeholder	Role
Citizens	Submit and track complaints
Municipal Officers	Manage and monitor complaints
Field Workers	Resolve assigned complaints
System Administrator	Manage the overall system
Municipal Departments	Receive and resolve complaints
Municipal Management	Monitor performance and reports

8. User Roles

Role	Description
Citizen	Can submit and track their own complaints.
Municipal Officer	Can manage complaints within their assigned municipal department.
Field Worker	Can view and resolve complaints assigned to them.
Administrator	Has system-level management privileges.

9. Functional Requirements

ID	Name	Requirement
FR-01	User Registration	The system shall allow citizens to create an account using name, email, phone number, address and password, with validation before account creation.
FR-02	User Login	The system shall authenticate registered users and redirect them to the dashboard corresponding to their role.
FR-03	Role-Based Access Control	The system shall provide different permissions for Citizen, Municipal Officer, Field Worker and Administrator.

ID	Name	Requirement
FR-04	Complaint Registration	Citizens shall submit title, description, category, location, priority where applicable, and supporting evidence. A unique complaint ID shall be generated.
FR-05	Complaint Categorization	The system shall support categories such as Roads, Garbage, Water Supply, Drainage, Electricity/Streetlights, Sanitation, Parks, Public Property, Environment and Other.
FR-06	Location Management	Citizens shall provide the problem location using a manual address, map-based location, or GPS-based location.
FR-07	Evidence Upload	Citizens shall upload photographs or other supported evidence associated with a complaint.
FR-08	Complaint ID Generation	The system shall automatically generate a unique complaint identifier.
FR-09	Complaint Verification	Municipal officers shall review complaints, verify or reject them, provide rejection reasons, and modify category or priority when necessary.
FR-10	Department Assignment	Verified complaints shall be assigned to an appropriate municipal department.
FR-11	Worker Assignment	Officers shall assign complaints to field workers with complaint ID, worker, department, priority, deadline and assignment date.
FR-12	Complaint Status Management	The system shall maintain statuses such as Submitted, Under Review, Verified, Assigned, In Progress, Resolved, Closed, Rejected, Reopened and Overdue.
FR-13	Complaint Tracking	Citizens shall track the current status and view a status timeline.
FR-14	Worker Progress Updates	Field workers shall update the progress of assigned complaints.
FR-15	Resolution Evidence	Workers shall upload evidence of completed work, preferably before/after images.
FR-16	Resolution Verification	Officers shall approve resolution, reject resolution, request additional work, or reopen the complaint.
FR-17	Complaint Reopening	Citizens may request reopening when an issue has not been properly resolved; officers review the request.
FR-18	Notifications	The system shall notify users about submission, verification, rejection, assignment, work start, resolution and reopening events.
FR-19	Feedback and Rating	Citizens shall provide a 1 5 rating and optional comments after resolution.
FR-20	Search and Filtering	Officers and administrators shall filter complaints by ID, category, status, priority, department, worker, location and date.
FR-21	Deadline Management	Officers shall specify resolution deadlines and the system shall identify complaints exceeding their deadlines.
FR-22	Escalation	The system shall identify overdue complaints and notify responsible officers; repeatedly overdue complaints may be escalated.
FR-23	Dashboard	The system shall provide role-specific dashboards for citizens, officers, workers and administrators.
FR-24	Reports	The system shall generate reports by category, department, location, status, priority, date, resolution time, worker performance and feedback.

10. AI-Based Functional Requirements

ID	Requirement
AI-FR-01: Automatic Complaint Classification	The system may analyze citizen-entered text and suggest a category/subcategory. Users and officers can modify the suggestion.
AI-FR-02: Priority Prediction	The system may analyze complaint information and suggest Critical, High, Medium or Low priority. The municipal officer retains final authority.
AI-FR-03: Duplicate Complaint Detection	The system may compare new complaints with existing complaints and identify potentially similar complaints. The officer decides whether complaints should be merged.

11. Non-Functional Requirements

ID	Requirement
NFR-01: Performance	Normal user requests should respond within approximately 3 seconds under normal load.
NFR-02: Security	Use secure authentication, password hashing, role-based authorization, input validation and protection of sensitive information.
NFR-03: Usability	Provide simple navigation, clear controls, easy complaint submission, clear status indicators and responsive design.
NFR-04: Reliability	Maintain complaint information reliably and prevent accidental data loss during normal operation.
NFR-05: Availability	The system should be available throughout the day except during planned maintenance.
NFR-06: Scalability	The system should support increasing numbers of citizens, complaints, departments, workers and officers.
NFR-07: Maintainability	Use modular architecture so new features, categories and departments can be added with minimal changes.
NFR-08: Compatibility	Support modern browsers such as Chrome, Edge and Firefox.
NFR-09: Data Privacy	Personal and complaint information shall only be accessible to authorized users.
NFR-10: Backup and Recovery	The database should be backed up periodically so important complaint information can be recovered after system failure.

12. External Interface Requirements

12.1 User Interface

Citizen: registration, login, dashboard, complaint submission, tracking, notifications and feedback.

Officer: dashboard, complaint management, assignment, reports and resolution verification.

Worker: dashboard, assigned complaints, details, progress updates and evidence upload.

Administrator: user, department and category management, dashboard and reports.

12.2 Software Interfaces

MongoDB database

Authentication system

File/image storage service

Mapping service

Notification service

12.3 Hardware Interfaces

Desktop computers

Laptops

Tablets

Smartphones

Smartphone camera for complaint evidence

13. Proposed Technology Stack

Layer	Technology
Frontend	React.js
Styling	CSS / Tailwind CSS
Backend	Node.js
Backend Framework	Express.js
Database	MongoDB
Database Management	MongoDB Compass
Authentication	JWT
Password Security	bcrypt
File Upload	Multer
File Storage	Cloudinary
Maps	Leaflet + OpenStreetMap
Charts	Recharts
API Testing	Postman
Version Control	Git + GitHub
Development Environment	VS Code

14. System Architecture

The proposed system follows a web-based client-server architecture. Citizens, municipal officers, field workers and administrators access the React frontend. Requests are processed through a REST API implemented using Node.js and Express.js. MongoDB stores application data, Cloudinary stores uploaded images, mapping services provide location support, and AI services provide classification, priority and duplicate-detection assistance.

High-level flow: Users React Frontend REST API Node.js + Express MongoDB / Cloudinary / Maps / AI Services.

15. Complaint Lifecycle

Citizen submits complaint.

Complaint ID is generated.

Complaint enters review.

Officer verifies complaint.

If invalid, the complaint is rejected and the citizen is notified.

If valid, a department and field worker are assigned.

Field worker performs the work and updates progress.

Resolution evidence is uploaded.

Officer verifies the resolution.

If unresolved, the complaint may be reopened and reassigned.

If resolved, feedback is collected and the complaint is closed.

16. Business Rules

A citizen must be authenticated before submitting a non-anonymous complaint.

Every complaint must have a unique complaint ID.

Every complaint must belong to a valid category.

Only authorized municipal officers can verify complaints.

Only authorized officers can assign complaints to workers.

A field worker can update only complaints assigned to them.

A worker should provide resolution evidence before marking a complaint completed.

Only an authorized officer can verify the final resolution.

Citizens can provide feedback only after the complaint has been resolved.

Citizens can request reopening after resolution.

AI-generated classifications and priorities are recommendations; officers retain final authority.

Users cannot access information or functionality outside their assigned role.

17. Assumptions

Citizens have access to the internet.

Users provide reasonably accurate information.

Municipal departments and workers are registered in the system.

Municipal officers are responsible for verifying complaints.

Field workers have access to smartphones or computers.

Internet connectivity is available during complaint processing.

Uploaded images comply with supported formats and size limits.

AI predictions are advisory and may require human verification.

18. Constraints

Internet connectivity is required.

AI prediction quality depends on available training data.

Map and cloud storage services may have usage limits.

Municipal staff must update complaint status accurately.

Large numbers of image uploads may increase storage requirements.

Some advanced features may require third-party APIs.

The initial project may be a prototype rather than a production municipal deployment.

19. Feasibility Analysis

19.1 Technical Feasibility

The system can be developed using widely available technologies such as React.js, Node.js, Express.js and MongoDB. Therefore, the project is technically feasible.

19.2 Economic Feasibility

The prototype can use open-source technologies and free/developer-tier services, keeping development cost low.

19.3 Operational Feasibility

The system is designed to simplify complaint registration for citizens and complaint management for municipal authorities. Therefore, it is operationally feasible.

19.4 Schedule Feasibility

The project can be developed in phases: requirement analysis, system design, database design, frontend development, backend development, integration, testing and deployment.

20. Security Requirements

- Secure authentication.

- Password hashing.

- JWT-based authorization.

- Role-based access control.

- Input validation.

- File type validation.

- File size restrictions.

- Protection against unauthorized API access.

- Secure handling of user information.

- Proper session/token management.

21. Testing Requirements

Unit Testing

Individual functions and modules shall be tested independently.

Integration Testing

Interactions between frontend/backend, backend/database and external services shall be tested.

System Testing

The complete complaint lifecycle shall be tested from submission through closure.

Security Testing

Unauthorized access and role permissions shall be tested.

Usability Testing

The system shall be evaluated for ease of complaint submission and tracking.

User Acceptance Testing

Representative municipal users shall evaluate whether the system meets specified requirements.

22. Success Criteria

Citizens can register and log in successfully.

Citizens can submit complaints.

Every complaint receives a unique ID.

Complaints can be categorized and assigned.

Officers can verify complaints.

Workers can receive assignments.

Workers can update progress.

Resolution evidence can be uploaded.

Officers can verify resolution.

Citizens can track complaints.

Citizens can provide feedback.

Overdue complaints can be identified.

Reports can be generated.

Role-based access works correctly.

AI modules provide useful recommendations without preventing normal complaint processing.

23. Future Enhancements

AI-powered image classification.

AI-powered multilingual complaint processing.

AI chatbot for citizens.

Voice-based complaint registration.

Advanced duplicate complaint detection.

Predictive maintenance.

Geographic complaint heat maps.

Mobile application for Android/iOS.

SMS and WhatsApp notifications.

Integration with existing government/municipal systems.

Advanced analytics and predictive dashboards.

Automated municipal resource allocation.

24. Conclusion

The Municipal Complaint Management System provides a centralized solution for reporting, managing, assigning, tracking and resolving municipal complaints. The system connects citizens with municipal authorities and field workers through a structured complaint lifecycle. It improves transparency through complaint tracking, resolution evidence, feedback and status history, while dashboards and reports help

authorities monitor performance and identify problem areas.

AI-assisted classification, priority prediction and duplicate detection can further improve efficiency while keeping final decisions under human supervision. The proposed system is technically, economically and operationally feasible and can serve as a foundation for a scalable smart-city civic service platform.